Clinical factors in the identification of small choroidal melanoma

Carol L. Shields, MD; Hakan Demirci, MD;
Miguel A. Materin, MD; Brian P. Marr, MD;
Arman Mashayekhi, MD; Jerry A. Shields, MD

ABSTRACT

The detection and treatment of choroidal melanoma early in its natural course is critical to providing the patient with the best prognosis. Studies of tumour doubling time have indicated that metastasis from choroidal melanoma can occur quite early in the course of the disease, when the tumour is about 3.0 mm in basal dimension and 1.5 mm in thickness. Clinical studies have shown that, at 5 years, metastasis occurs in 16% of patients with small choroidal melanomas (less than 4 mm thick), compared with 32% of those with medium-sized (4–8 mm thick) choroidal melanomas and 53% of those with large (more than 8 mm thick) choroidal melanomas. The difficulty with early detection of choroidal melanoma relates to its clinical similarity to benign choroidal nevus. Factors that assist in differentiating small choroidal melanoma from choroidal nevus can be remembered using the mnemonic "TFSOM" (to *f*ind *s*mall *o*cular *m*elanoma), where T = thickness greater than 2 mm, F = subretinal fluid, S = symptoms, O = orange pigment and M = margin touching optic disc. Choroidal melanocytic tumours that display none of these factors have a 3% risk of growth into melanoma at 5 years and most likely represent choroidal nevi. Tumours that display one factor have a 38% risk of growth, and those with two or more factors show growth in over 50% of cases. Most tumours with two or more risk factors probably represent small choroidal melanomas, and early treatment is generally indicated. Therefore, ophthalmologists should be aware of the clinical factors that identify small choroidal melanoma so that early treatment and better prognosis can be achieved for their patients.

There is increasing interest in early detection and intervention for human cancers. We have learned much about the behaviour of uveal melanoma from the literature on management of cutaneous melanoma. Detection and treatment of cutaneous melanoma at an early stage have been correlated with improved patient survival, and this observation has been especially notable during the past 40 years.[1–5] Cutaneous melanoma is now recognized when the tumour is thinner and less invasive than in the past.[3] In addition, cutaneous melanoma now tends to be treated when the tumour is in a radial growth phase (superficial spreading) and is less likely to be ulcerated than in the past.[3] Dermatologists have taught clinicians to identify early cutaneous melanoma by using the mnemonic "ABCD": *a*symmetry, *b*order irregularity, *c*olour variegation and *d*iameter larger than a pencil eraser.[5] Pigmented skin lesions with one or more of these features should be excised and inspected histopathologically for cutaneous melanoma. With this approach, cutaneous melanoma is most often detected at an early stage.

From the Oncology Service, Wills Eye Hospital, Thomas Jefferson University, Philadelphia, Pa.

Correspondence to: Dr. Carol L. Shields, Oncology Service, Wills Eye Hospital, 840 Walnut St., Philadelphia PA 19107, USA; fax (215) 928-1140, carol.shields@shieldsoncology.com

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